CARGO DWELL TIME REDUCTION

Port of Zeebrugge — Executive Summary

Business Analysis Case · Philippe Godfroy · Q4 2024 · CONFIDENTIAL

BUSINESS ANALYSIS
Logistics · Process Improvement

01 — PROBLEM STATEMENT

Container terminals at the Port of Zeebrugge are experiencing an average dwell time of **6.4 days** per container — **38% above the sector benchmark of 4.6 days**. This results in significant demurrage costs estimated at **€2.1M annually**, reduced terminal throughput capacity, and declining service levels for shippers and shipping lines. This analysis identifies the root causes and proposes targeted interventions to reduce average dwell time by a minimum of 25%.

6.4d	43%	4.2h	€2.1M
Avg. Dwell Time	Containers > 5 days	Avg. Doc Processing	Annual Demurrage
▲ 38% vs sector	▲ High risk	Root cause #1	▲ YoY increase

02 — KEY FINDINGS

43%	Tue/Wed	3 agents	4.2h → 1.1h
Documentation delays	**Peak congestion days**	**No EDI capability**	**Doc time reduction**
Primary root cause	Shipper deadline clustering	Evergreen, MSC, CMA CGM	Via automated B/L matching

Quantitative analysis of 500 container movements (Jan 2023 – Jun 2024) revealed that documentation-related delays are the dominant driver of excess dwell time, accounting for 43% of all containers exceeding the 5-day threshold. A secondary factor is demand clustering: 61% of weekly container arrivals occur on Tuesday and Wednesday, overwhelming customs pre-clearance capacity and causing systemic backlogs. Three of five major shipping agents lack EDI integration, forcing manual document handling that averages 4.2 hours per container.

03 — ROOT CAUSE BREAKDOWN

Delay Reason	% of Containers >5d	Avg. Doc Hours	Est. Annual Cost	Addressable via EDI?
Documentation delay	43%	4.2h	€ 903,000	YES
Customs clearance	22%	2.8h	€ 462,000	PARTIAL
Port congestion	15%	—	€ 315,000	NO
Late shipper pickup	12%	—	€ 252,000	NO
Inspection required	8%	—	€ 168,000	NO

04 — RECOMMENDATIONS

HIGH PRIORITY	**Implement EDI integration with Evergreen Europe & MSC Zeebrugge** Est. impact: –1.8 days avg. dwell time · Implementation: Q1 2025 · Cost: €85,000
HIGH PRIORITY	**Automate B/L matching & pre-clearance notification workflow** Reduces doc processing from 4.2h → 1.1h · ROI positive within 4 months
MEDIUM PRIORITY	**Redistribute import arrival scheduling to balance Tue/Wed peak load** Coordinate with top 5 shippers · Est. –0.4 days dwell · Zero cost
LOW PRIORITY	**Deploy mobile supervisor dashboard for real-time dwell time monitoring** Enables proactive intervention · Development: 6 weeks · Cost: €12,000

05 — EXPECTED OUTCOMES (POST-IMPLEMENTATION)

4.8d	**22%**	**1.1h**	**€840K**
Target Dwell Time	Containers > 5 days	Doc Processing Time	Annual Net Saving
▼ –25% reduction	▼ –21 percentage pts	▼ via automation	▲ net of invest.

Methodology: Analysis based on 500 container movement records (Jan 2023 – Jun 2024), supplemented by stakeholder interviews with terminal operations (2), customs liaison (1), and shipping agent representatives (3). Process mapping conducted using BPMN 2.0. Financial impact estimates derived from published Port of Zeebrugge demurrage tariffs and internal cost benchmarks. ROI calculations assume 80% EDI adoption rate within 12 months.